

Constructible Polygons

An ancient problem and its solution

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They defined *constructible* to mean that something could be drawn using only a compass and a straightedge.

A number x is considered constructible if a line segment with length x can be constructed with compass and straightedge.

But how do we tell whether something is constructible?

The Algebraic Sense

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Theorem: Constructibility

A real number x is constructible if and only if x can be written using the operations $+$, $-$, $\cdot$, $/$, $\sqrt{}$ finitely many times on integers.

A complex number $z = x + iy$ is constructible if and only if the real (x) and imaginary (y) parts of z are constructible.

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Corollary

The constructible numbers form a Euclidean field.

Criterion

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Definition: Regular Polygon

A polygon is considered *regular* if all sides are of the same length and all angles are the same measure.

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Proposition: Polygon Constructibility

A regular polygon with n sides (a regular n -gon) is constructible if and only if $\cos\left(\frac{2\pi}{n}\right)$ is constructible.

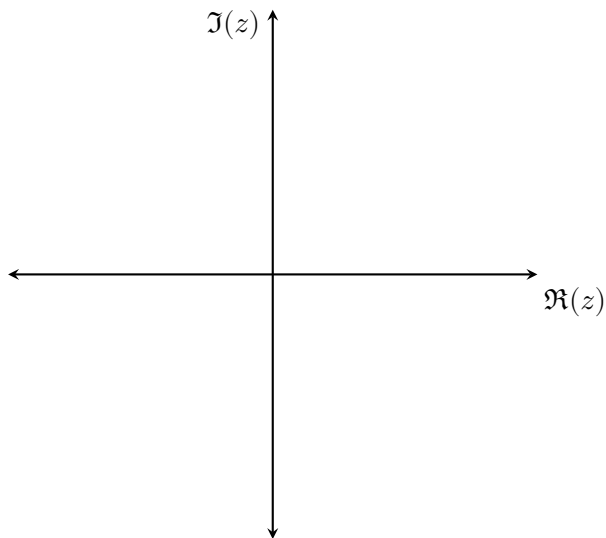
Proof of Proposition

Proof. Consider a regular n -gon inscribed in a circle of radius r (WLoG $r = 1$). We further insist WLoG that a vertex of the n -gon lies at the point $(1, 0)$ or the complex number $z = 1$. Then the other vertices represent the n^{th} roots of unity, with the first representing $\zeta_n = e^{\frac{2\pi i}{n}}$. By Euler's formula, the real and imaginary parts of ζ_n are $\cos\left(\frac{2\pi}{n}\right)$ and $\sin\left(\frac{2\pi}{n}\right)$. We can write

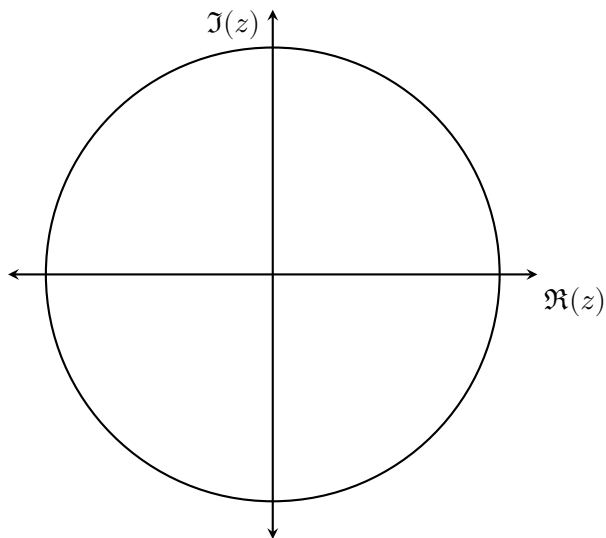
$$\sin\left(\frac{2\pi}{n}\right) = \sqrt{1 - \cos^2\left(\frac{2\pi}{n}\right)}$$

so ζ_n is constructible if and only if $\cos\left(\frac{2\pi}{n}\right)$ is constructible.

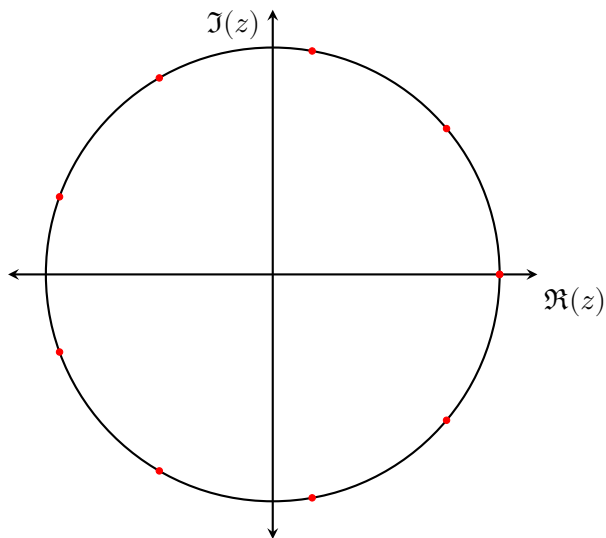
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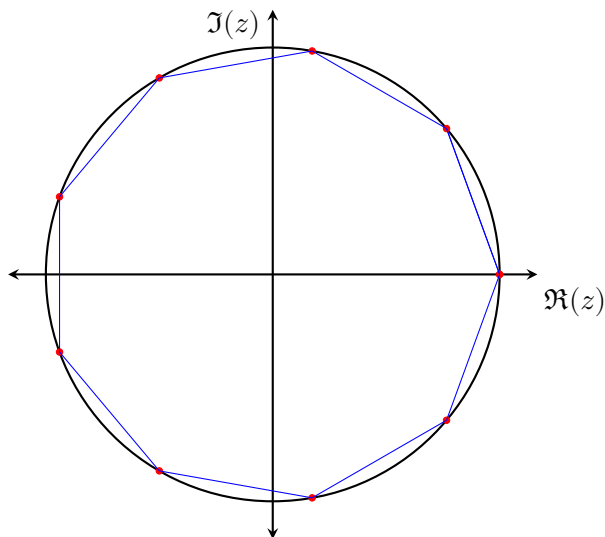
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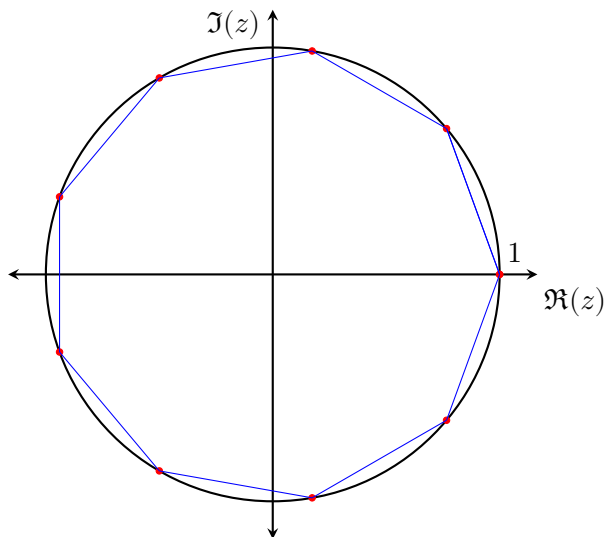
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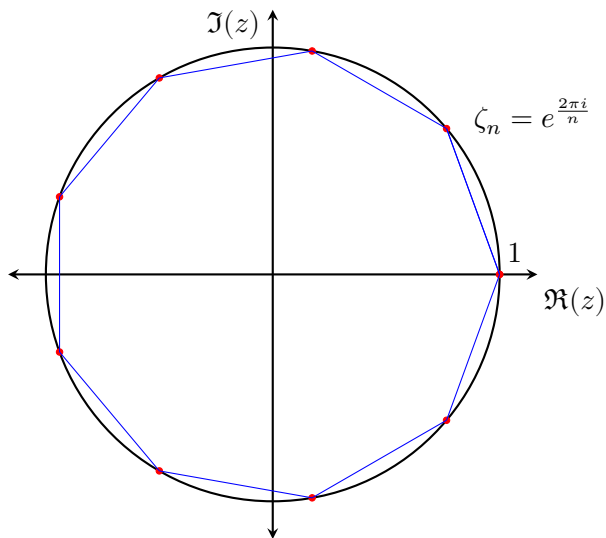
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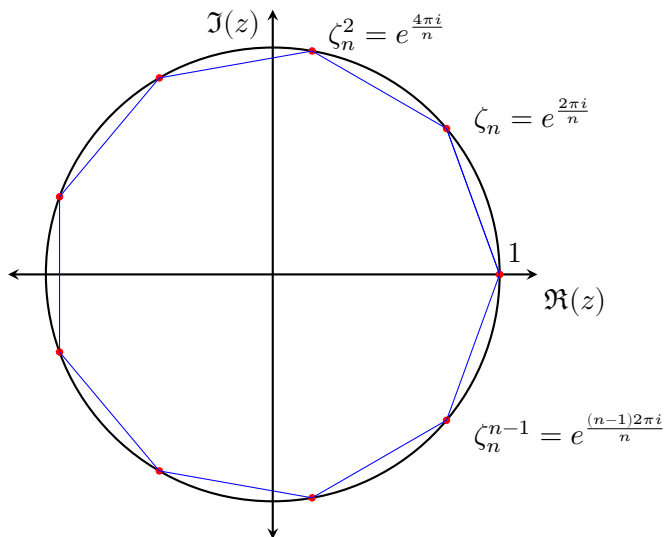
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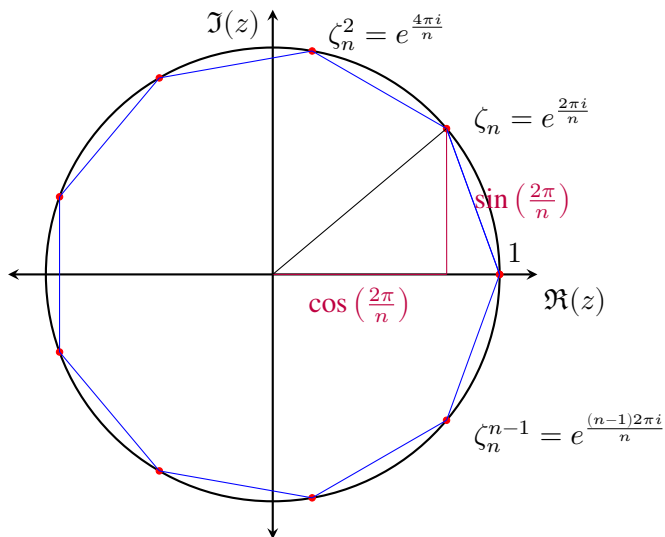
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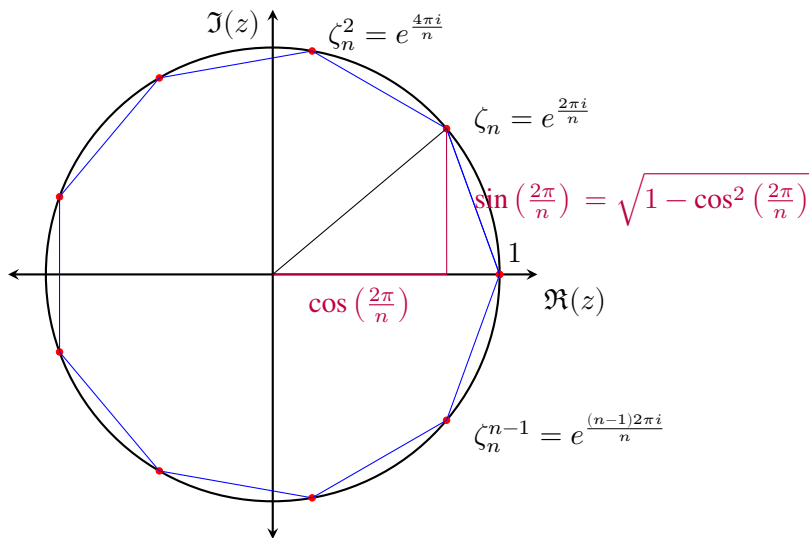
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$$\begin{aligned}\cos\left(\frac{2\pi}{17}\right) &= \frac{1}{16} \left(\sqrt{17} - 1 + \sqrt{34 - 2\sqrt{17}} \right) \\ &\quad + \frac{1}{8} \left(\sqrt{17 + 3\sqrt{17} - \sqrt{34 - 2\sqrt{17}} - 2\sqrt{34 + 2\sqrt{17}}} \right).\end{aligned}$$

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This is hard to do, and it also doesn't prove that polygons *can't* be constructed.

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Definition: Field Extensions

A field extension $K/\mathbf{Q}$ is a field K that contains $\mathbf{Q}$. Such extensions can be thought of as *vector spaces* over $\mathbf{Q}$.

What are Cyclotomic Fields?

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Definition: Cyclotomic Field

A Cyclotomic Field is an extension of the rational numbers,

$$\mathbf{Q}(\zeta_n) = \{a_0 + a_1\zeta_n + a_2\zeta_n^2 + \cdots + a_{n-1}\zeta_n^{n-1} : a_i \in \mathbf{Q}\}$$

where ζ_n is a primitive n^{th} root of unity.

The Basic Idea

We want to have an idea of the “size” and relationship of $\mathbf{Q}(\zeta_n)$ with respect to $\mathbf{Q}$.

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One way to do this is to "scramble" the field $\mathbf{Q}(\zeta_n)$ while keeping $\mathbf{Q}$ fixed.

Since every element of $\mathbf{Q}(\zeta_n)$ can be expressed as a linear combination of powers of ζ_n , we can reduce this to simply permuting the generators, ζ_n^m for $0 \leq m \leq n$.

The Galois Group of Cyclotomic Fields

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Definition: The Galois Group

The Galois Group of a field extension $\mathbf{Q} \subseteq K$, denoted $\text{Gal}(K/\mathbf{Q})$, is the group of automorphisms (isomorphisms from K to K) keeping $\mathbf{Q}$ fixed.

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Theorem: Galois Group of a Cyclotomic Field

The Galois Group of the n^{th} Cyclotomic Field $\mathbf{Q}(\zeta_n)$, denoted $\text{Gal}(\mathbf{Q}(\zeta_n)/\mathbf{Q})$, is isomorphic to the group of units mod n :

$$\text{Gal}(\mathbf{Q}(\zeta_n)/\mathbf{Q}) \cong U_n$$

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However, when we do so, we want to send ζ_n to another primitive n^{th} root of unity, since these generate the total group of n^{th} roots of unity.

We can see that the only way to do so is if $(i, n) = 1$. This means that $i \in U_n$. Since $(a^k)^j = a^{kj}$, we can see that the Galois group is the same as U_n .

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$$\cos\left(\frac{2\pi}{n}\right) = \frac{e^{\frac{2\pi i}{n}} + e^{-\frac{2\pi i}{n}}}{2} = \frac{\zeta_n + \zeta_n^{-1}}{2}$$

since

$$\zeta_n = e^{\frac{2\pi i}{n}}.$$

Why Does the Galois Group Matter?

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Theorem: Constructible Numbers

A number $x \in \mathbf{C}$ is constructible if and only if it is contained in an extension K of $\mathbf{Q}$ such that

$$|\mathrm{Gal}(K/\mathbf{Q})| = 2^k$$

for some $k \in \mathbf{N}$.

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From the constructibility theorem, we need x to be writeable using only the operations $+$, $-$, $\cdot$, $/$ as well as square roots.

Thus, it is essential that x is contained in such an extension, where every number added is (more or less) the square root of some other number.

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But we already know this!

Redux: The Group U_n

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where φ is Euler's totient function.

This is given by

$$\varphi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right).$$

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This gives us the following theorem:

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If $p \neq 2$ and $p^2 \mid n$, we can see that $p \mid \varphi(n)$.

This gives us the following theorem:

Theorem: Gauss-Wantzel

A regular n -gon is constructible if and only if

$$n = 2^k p_1 \cdots p_n$$

where $k \geq 0$ and p_i are distinct Fermat primes.

Fermat Primes

Definition: Fermat Numbers

Recall that Fermat Numbers are numbers of the form

$$F_n = 2^{2^n} + 1.$$

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Conjecture: Fermat Primes

F_n is composite for all $n > 4$.

The Constructible n -gons

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The only known Fermat primes are:

$$F_0 = 3 \quad F_1 = 5 \quad F_2 = 17 \quad F_3 = 257 \quad F_4 = 65537$$

Taking any combination of these, along with powers of 2, produces the constructible regular n -gons.

The powers of 2 tells us that there are infinitely many constructible regular n -gons.

The Constructible n -gons

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The first few constructible n -gons are:

3, 4, 5, 6, 8, 10, 12, 15, 16, 17, 20, 24, 30, 32, 34, 40, 48, 51, 60, 64, 68, 80, ...

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The first few constructible n -gons are:

3, 4, 5, 6, 8, 10, 12, 15, 16, 17, 20, 24, 30, 32, 34, 40, 48, 51, 60, 64, 68, 80, ...

The first few non-constructible n -gons are:

7, 9, 11, 13, 14, 18, 19, 21, 22, 23, 25, 26, 27, 28, 29, 31, 33, 35, 36, 37, 38, ...

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The main ideas used here, from both constructibility and Galois Theory, can be used to solve many such problems from antiquity:

1. Squaring the circle (π is transcendental, and thus non-constructible)
2. Doubling the cube ($\sqrt[3]{2}$ is non-constructible)
3. Trisecting the angle (Ex. $\cos\left(\frac{\pi}{9}\right)$ is non-constructible)

Thank you!

Any questions?